Predicting Loan Default Risk at LendingClub

Presented By :

Team 9:

Chamnan Suon
Daaksha Arun
Parthvi Joshi
Quency Ye

Table of Content

1. Introduction

Problem Statement

Lenders/Investors
Individuals,
bank, financial institutions

Lending Club
Launched 2007
Peer-to-peer lending marketplace

Borrowers
Individual or small businesses
50 US states
2.3 millions accepted

Loan Default
Borrowers fail to make payments

Problem
Loss of money
Loss of trust
Revenue and operational impact

2. Data Cleaning & Pipeline Preprocessing

Data Source and Cleaning

Data Source and Description

- Kaggle
- Data between 2007 and 2018
- 50 US States
- Data size 1.68GB
- 151 Columns and 2.2 million rows

Data Cleaning

- Drop unused columns
- Convert columns types
- Derived new columns (issue_month, issue_year, loan_target)
- Filter out records from Maine, Vermont, New Hampshire, Massachusetts, Connecticut, Rhode Island, New York

Feature Selection and Preprocessing

Numeric

1. loan amount
2. funded amount
3. interest rate
4. installment
5. annual income
6. debt to income ratio
7. fico range low
8. fico range high
9. number of open credit lines
10. public record of derogatory
11. credit revolving balance
12. credit revolving utilization
13. total credit lines
14. month loan funded
15. year loan funded

impute with mean

standardize numeric features

Feature Selection and Preprocessing

Categorical

16. term
17. grade
18. sub grade
19. employment length
20. home ownership
21. verification status
22. purpose
23. address state
24. application type
25. loan_target (target feature)

impute with mode

use one-hot encoder

3. Exploratory Data Analysis

Correlation Matrix Between Features

- Most features show weak correlation with the default
- Interest rates are heavily based on loan grades
- Larger loans have higher monthly payments
- Lower credit scores predict higher defaults

Correlation Matrix (Numeric + Encoded Categorical Features)

	loan_amnt	int_rate	installment	annual_inc	dti	fico_range_low	open_acc	pub_rec	revol_bal	revol_util	total_acc	issue_year	issue_month	grade_cat	sub_grade_cat	emp_length_cat	home_ownership_cat	verification_status_cat	purpose_cat	term_cat	application_type_cat	loan_target
loan_amnt	1.00	0.10	0.95	0.19	0.04	0.11	0.18	-0.06	0.32	0.10	0.20	0.06	-0.01	0.10	0.10	0.01	-0.18	0.20	-0.15	0.40	0.12	0.02
int_rate	0.10	1.00	0.12	-0.05	0.13	-0.42	-0.01	0.05	-0.03	0.26	-0.04	-0.05	0.01	0.95	0.98	0.00	0.07	0.25	0.09	0.37	0.03	0.21
installment	0.95	0.12	1.00	0.18	0.05	0.05	0.17	-0.05	0.31	0.12	0.18	0.05	-0.01	0.12	0.12	0.01	-0.15	0.21	-0.14	0.14	0.10	0.03
annual_inc	0.19	-0.05	0.18	1.00	-0.08	0.03	0.09	-0.00	0.18	0.03	0.11	0.02	0.00	-0.05	-0.05	0.00	-0.09	0.02	0.00	0.04	-0.04	-0.02
dti	0.04	0.13	0.05	-0.08	1.00	-0.03	0.18	-0.03	0.10	0.11	0.14	0.07	0.01	0.12	0.12	-0.02	-0.02	0.07	-0.05	0.06	0.22	0.04
fico_range_low	0.11	-0.42	0.05	0.03	-0.03	1.00	0.02	-0.19	0.01	-0.47	0.01	0.08	0.01	-0.42	-0.44	-0.01	-0.10	-0.15	0.06	0.01	0.09	-0.12
open_acc	0.18	-0.01	0.17	0.09	0.18	0.02	1.00	-0.02	0.24	-0.13	0.71	0.03	-0.01	-0.00	-0.01	-0.00	-0.14	0.02	-0.07	0.08	-0.02	0.02
pub_rec	-0.06	0.05	-0.05	-0.00	-0.03	-0.19	-0.02	1.00	-0.09	-0.05	0.00	0.01	-0.02	0.06	0.06	-0.02	0.01	0.03	0.01	-0.02	-0.02	0.03
revol_bal	0.32	-0.03	0.31	0.18	0.10	0.01	0.24	-0.09	1.00	0.25	0.21	-0.00	-0.01	-0.03	-0.03	-0.02	-0.16	0.07	-0.08	0.09	0.00	-0.02
revol_util	0.10	0.26	0.12	0.03	0.11	-0.47	-0.13	-0.05	0.25	1.00	-0.09	-0.16	-0.02	0.27	0.28	0.01	-0.03	0.13	-0.12	0.07	-0.03	0.06
total_acc	0.20	-0.04	0.18	0.11	0.14	0.01	0.71	0.00	0.21	-0.09	1.00	-0.06	-0.02	-0.03	-0.03	-0.03	-0.21	0.03	-0.05	0.10	-0.03	0.01
issue_year	0.06	-0.05	0.05	0.02	0.07	0.08	0.03	0.01	-0.00	-0.16	-0.06	1.00	-0.05	-0.10	-0.10	-0.04	0.00	-0.10	0.01	0.02	0.22	-0.13
issue_month	-0.01	0.01	-0.01	0.00	0.01	0.01	-0.01	-0.02	-0.01	-0.02	-0.02	-0.05	1.00	-0.01	-0.01	0.01	0.00	-0.01	-0.00	0.01	0.03	-0.03
grade_cat	0.10	0.95	0.12	-0.05	0.12	-0.42	-0.00	0.06	-0.03	0.27	-0.03	-0.10	-0.01	1.00	0.97	0.00	0.07	0.24	0.10	0.38	0.00	0.23
sub_grade_cat	0.10	0.98	0.12	-0.05	0.12	-0.44	-0.01	0.06	-0.03	0.28	-0.03	-0.10	-0.01	0.97	1.00	0.00	0.07	0.26	0.10	0.39	0.01	0.23
emp_length_cat	0.01	0.00	0.01	0.00	-0.02	-0.01	-0.00	-0.02	-0.02	0.01	-0.03	-0.04	0.01	0.00	0.00	1.00	0.06	-0.04	-0.00	-0.00	0.02	0.00
home_ownership_cat	-0.18	0.07	-0.15	-0.09	-0.02	-0.10	-0.14	0.01	-0.16	-0.03	-0.21	0.00	0.00	0.07	0.07	0.06	1.00	-0.02	0.03	-0.11	-0.08	0.05
verification_status_cat	0.20	0.25	0.21	0.02	0.07	-0.15	0.02	0.03	0.07	0.13	0.03	-0.10	-0.01	0.24	0.26	-0.04	-0.02	1.00	0.03	0.10	0.01	0.09
purpose_cat	-0.15	0.09	-0.14	0.00	-0.05	0.06	-0.07	0.01	-0.08	-0.12	-0.05	0.01	-0.00	0.10	0.10	-0.00	0.03	0.03	1.00	-0.07	0.00	0.01
term_cat	0.40	0.37	0.14	0.04	0.06	0.01	0.08	-0.02	0.09	0.07	0.10	0.02	0.01	0.38	0.39	-0.00	-0.11	0.10	-0.07	1.00	0.07	0.09
application_type_cat	0.12	0.03	0.10	-0.04	0.22	0.09	-0.02	-0.02	0.00	-0.03	-0.03	0.22	0.03	0.00	0.01	0.02	-0.08	0.01	0.00	0.07	1.00	-0.04
loan_target	0.02	0.21	0.03	-0.02	0.04	-0.12	0.02	0.03	-0.02	0.06	0.01	-0.13	-0.03	0.23	0.23	0.00	0.05	0.09	0.01	0.09	-0.04	1.00

Borrowers Characteristics

FICO score

Poor (<579)	0.00%
Fair (580-669)	16.38%
Good (670-739)	**71.26%**
Very Good (740-799)	11.01%
Exceptional (>800)	1.36%

Annual Income

<40k	18.33%
40-70k	**38.20%**
70-100k	23.81%
100-150k	13.85%
150-300k	5.18%
>300k	0.61%

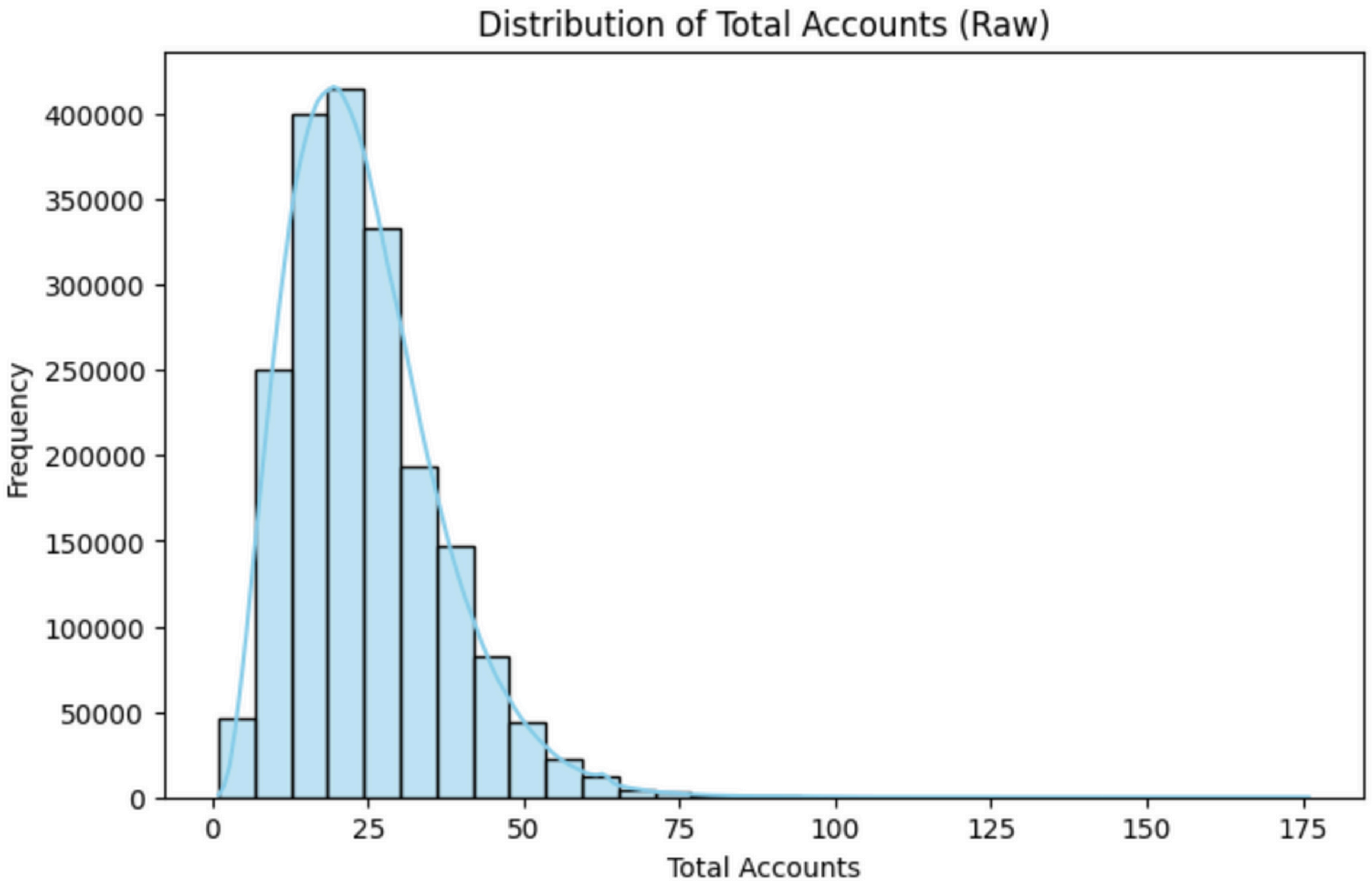
Employment Group

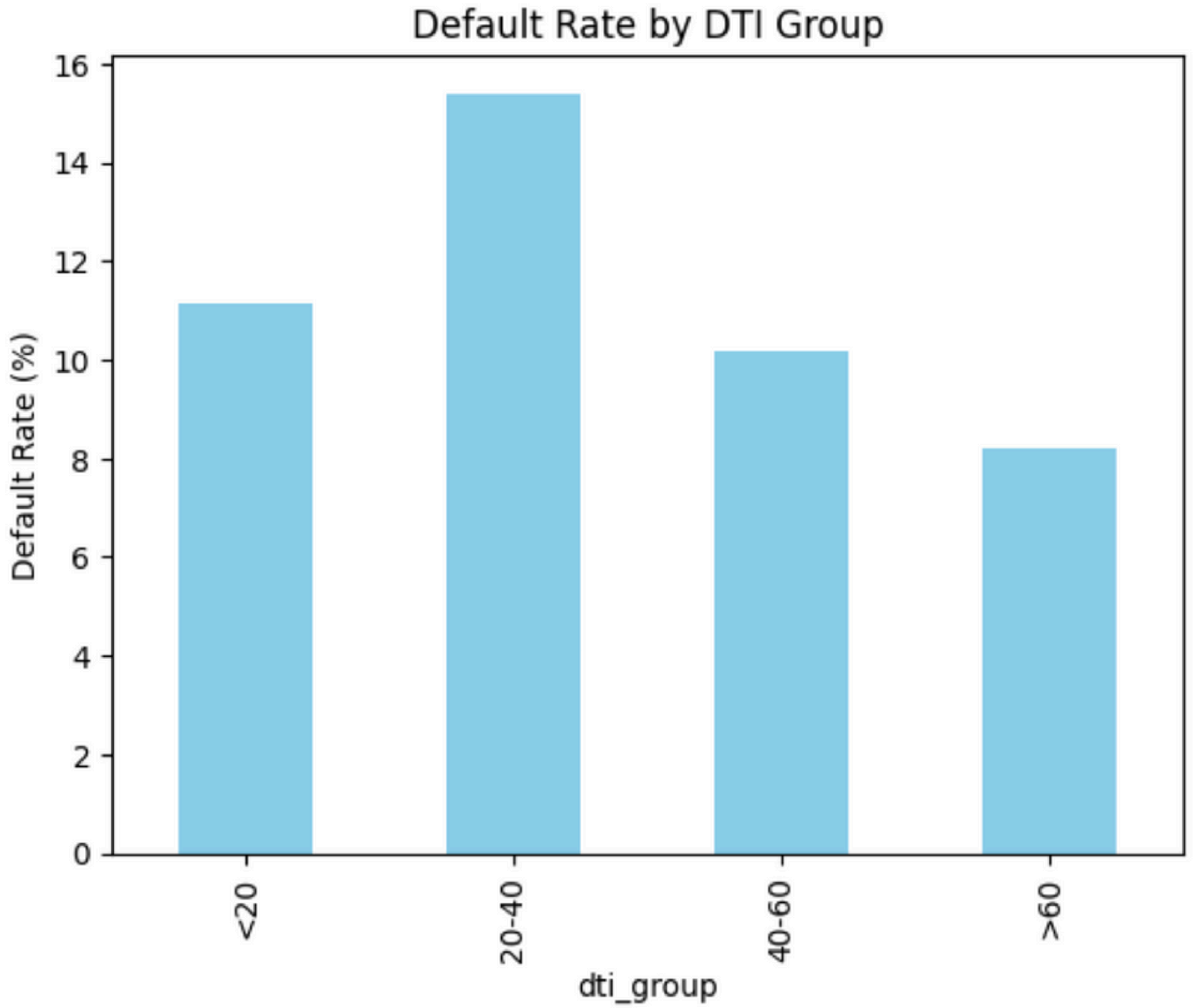
New (0-1)	16.08%
Early(2-3)	18.18%
Mid(4-6)	17.94%
Senior(7-9)	12.5%
Very Senior(10+)	**35.30%**

Mortgage account

0	**58.37 %**
1-2	25.99%
3-4	1.15%
5-9	4.31%
10+	0.19%

Debt and Income Ratios(DTI)

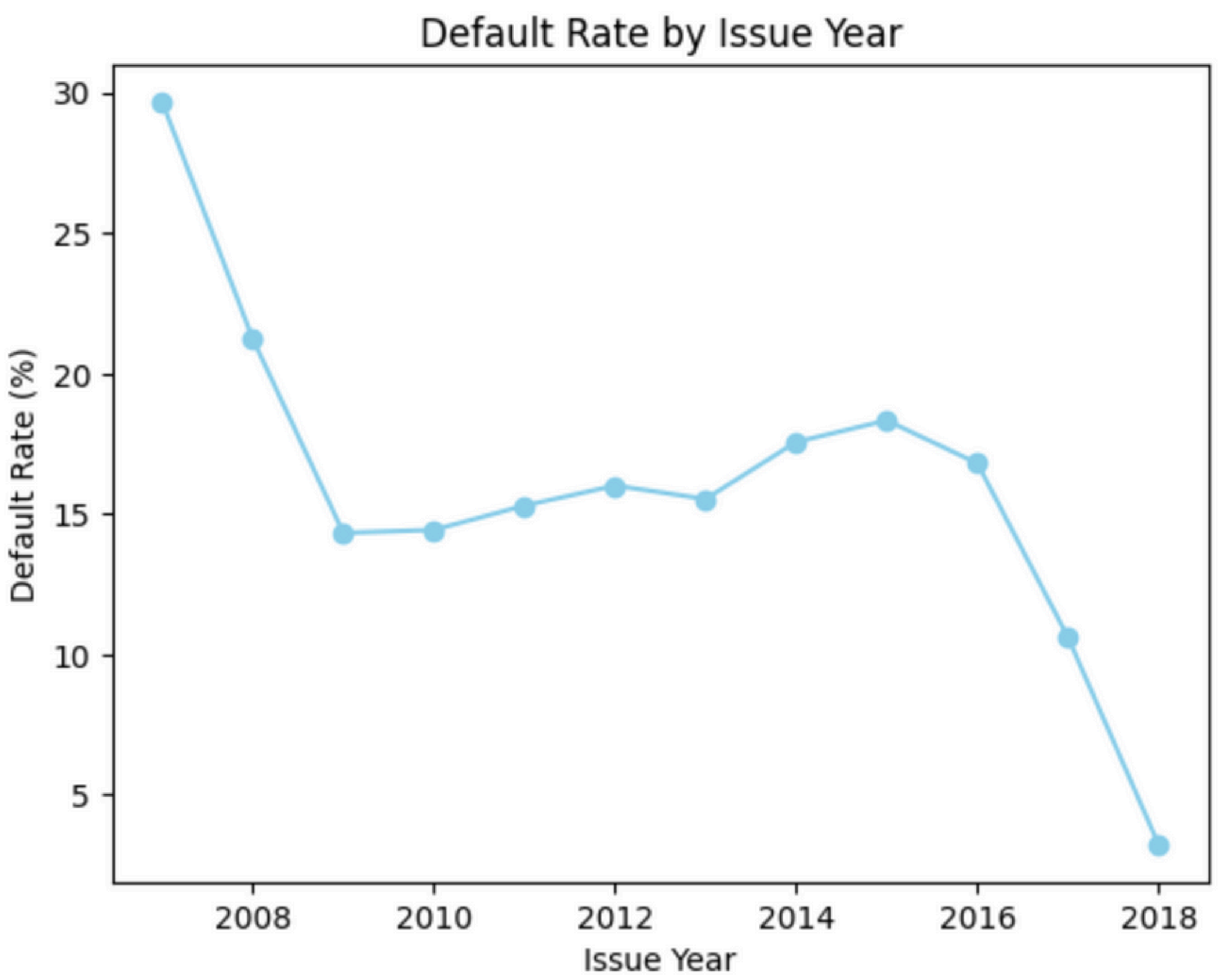


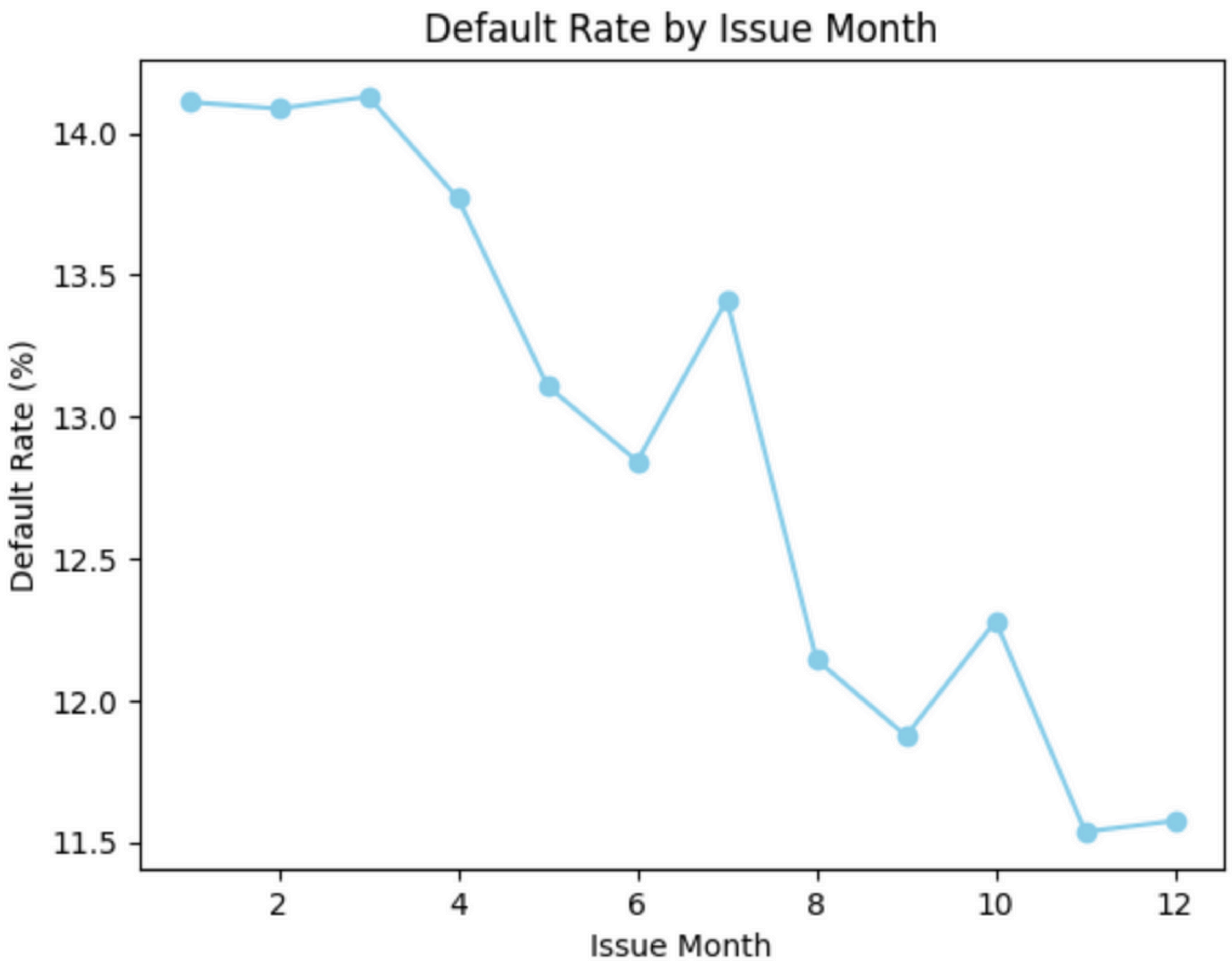


- The distribution of total accounts is right-skewed, with most borrowers holding between 10 and 30 accounts.

- DTI（20-40）--> higher default rate

Default Performance





- Default rates were highest in the early years (around 2007–2008) and gradually declined over time
- Monthly defaults are relatively stable, with a slight decrease toward the end of the year

4. Models

Model Selection : Random Forest

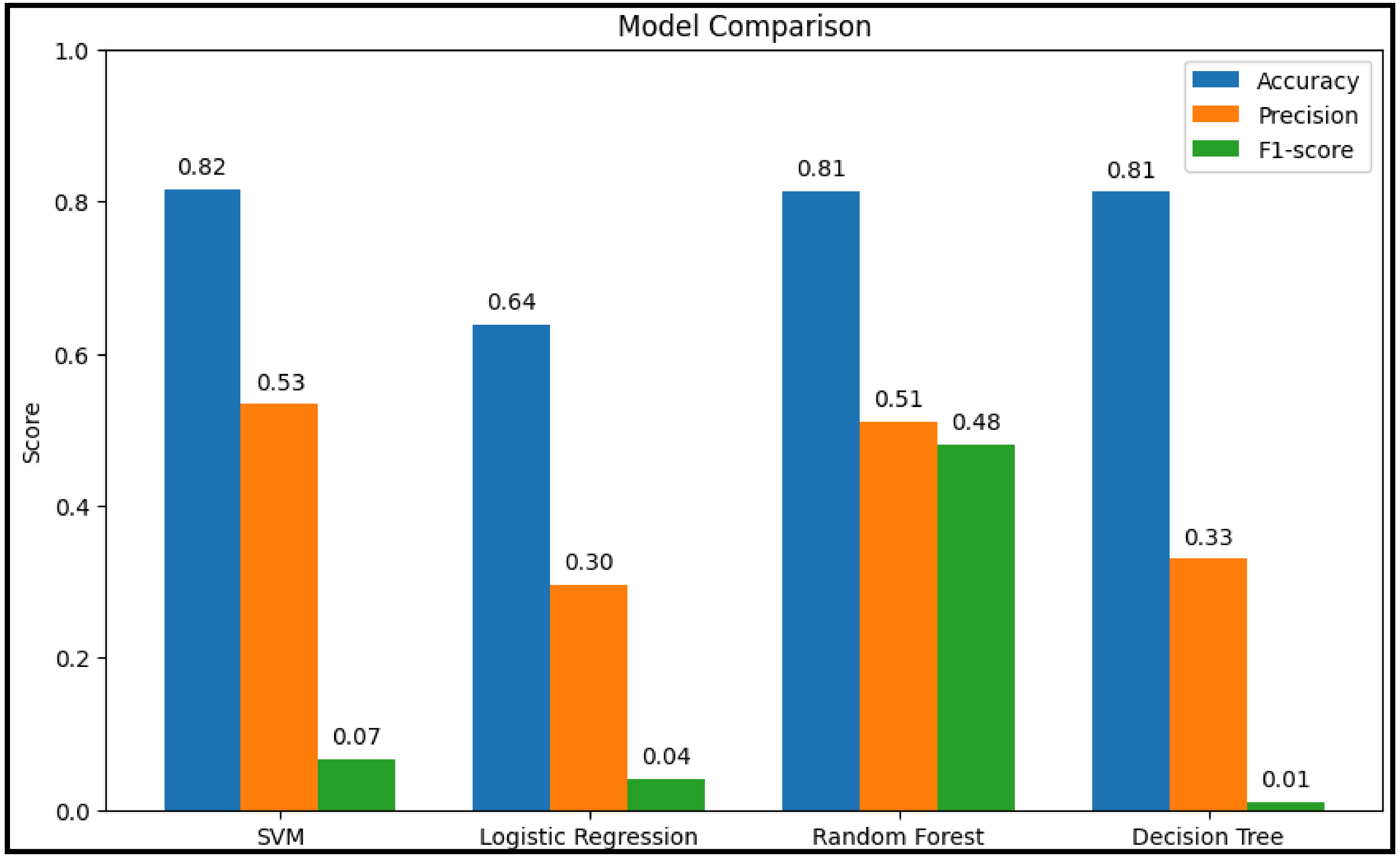
The Winner is Random Forest!

- Borrowers with **higher interest rates (~15.7%)**, **lower FICO scores, and high revolving credit utilization** were significantly more **likely to default**, showing strong pricing, credit quality, and financial stress effects.

- A key non-linear insight captured by Random Forest is that borrowers with **mid-range DTI (20–40%)** had the **highest default risk**, even higher than very high DTI borrowers.

- **60-month loans** consistently showed **higher interest rates** and higher default risk compared to **36-month loans,** highlighting the risk of longer loan maturities.

Comparison of Models



- Accuracy is similar (~0.81) across majority of the models.
- SVM & Logistic Regression show very low F1-scores → poor detection of loan default.
- Random Forest performs best overall, with the highest F1 and balanced precision → selected model.

Limitation and Challenges

- Limited time restricted us from testing advanced models like XGBoost , Ensemble methods and others which could potentially improve performance.
- The dataset is from 2018, so more recent data could provide better accuracy and reflect current patterns.
- SVM took nearly 3 hours to run due to high dimensionality and computational cost.
- Additional feature engineering and optimization could strengthen model performance but was limited by time.

5. Conclusion

Conclusion

- Financial institutions can integrate different prediction models into their credit evaluation process to enhance risk assessment

- Continuous monitoring of monthly and yearly default patterns can support early-warning systems for economic downturns or policy changes.

- From a portfolio management perspective, institutions should use model outputs to support loan segmentation and targeted risk control

Recommendations for Future Work

- Explore advanced ensemble methods or combine predictions from multiple models, which typically provide the best performance for credit risk prediction

- Expanding the dataset with recent data. The most recent data reflect current economic conditions, lending practices, or borrower behavior

- Optimize computational efficiency. Use important features to eliminate features contributing <1% to predict

Link to Notebook

Google Colab
google.com

References

1. Kaggle dataset - https://www.kaggle.com/datasets/wordsforthewise/lending-club/data
2. LendingClub - https://www.lendingclub.com/

Thank You!

Questions?